

Biology, 2017 EC, entrance pilot test, booklet 06

Grade 12 · Biology · 2017 E.C.

Time allowed: 120 minutes

Questions: 38

Total marks: 38

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is correct about negative feedback mechanism in animals? It keeps the normal set point by (1)
 - A. coordinating sensory, control center and effector components to return the change back.
 - B. modifying the anatomical structures involved in the specific physiological activities.
 - C. initiating the effector components of the organisms to accelerate a change in the body's condition.
 - D. randomly depending on the type of stimulus that brings change to the internal environment.
2. Enzyme 'A' catalyzes only substrate 'B'. This indicates that enzymes are (1)
 - A. specific to their substrates.
 - B. sensitive to their substrates.
 - C. agents that reduce activation energy.
 - D. components of the substrate molecules.
3. Which one of the following statements is correct about the tertiary structure of a protein? It is determined by (1)
 - A. the arrangement of the alpha and beta sheets by hydrogen bonding.
 - B. the amino acid sequence of the polypeptide linked together by peptide bond.
 - C. the hydrogen bond, ionic bond and hydrophobic interactions that results in a globular shape.
 - D. the arrangement of multiple protein subunits associated in several protein chains.
4. Which of the following is correct about lock and key model of an enzyme action? During enzyme-substrate binding (1)
 - A. active site forms covalent interaction when it filled with substrate.
 - B. substrate binds on an enzyme to form permanent chemical bonding.
 - C. the shape of an enzyme changed slightly as substrate binds to form product.
 - D. substrate binds on an enzyme's active site and forms enzyme substrate-complex.
5. The presence of one copy of the Y-chromosome in human individual gives an evidence that the chance of getting (1)
 - A. male offspring is determined by mothers.
 - B. female offspring is determined by mothers.
 - C. female offspring is determined by fathers.
 - D. male offspring is determined by fathers.
6. This item is based on Mendelian and Non-Mendelian inheritance listed below. (1)
 - I. Co-dominance is observed.
 - II. Dominance of one allele is present.
 - III. Inheritance is determined by single gene.
 - IV. Inheritance is determined by multiple genes.
 - V. Offspring phenotypic ratio can be predictable.

Based on the above list, which numbers contain the characteristics that only belong to Mendelian inheritance?

 - A. I, III and IV
 - B. II, III and V
 - C. II, IV and V
 - D. I, II and V

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7. Genetic pedigree analysis enables the scientists to study (1)
- A. how traits and disease are inherited.
 - B. the effects of environment on phenotypes.
 - C. how genetic traits prevent human from infectious disease.
 - D. morphological variation between individuals.
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8. Which one of the following alternatives contains a list of common human genetic disorders? (1)
- A. Down syndrome, Cancer and Tuberculosis
 - B. Albinism, Diabetes and Schizophrenia
 - C. Hemophilia, Cystic fibrosis and Albinism
 - D. Diabetes, Leprosy and Down syndrome
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9. Which of the following is the correct definition of bioinformatics? It is the (1)
- A. conservation of biological diversity.
 - B. utilization of biotechnology for data dissemination.
 - C. storage, organization and retrieval of biological data.
 - D. science of study about information technology and its use.
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10. Musculoskeletal systems as the combination of muscular and skeletal systems, (1)
- A. involve in the transport of materials from heart to other body parts.
 - B. provide shape, form and protect vital organ from damage.
 - C. responsible for gas exchange between inside and outside environment.
 - D. cover the outer body surface and serves as barrier between the inside and outside environment.
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11. How does the contraction of skeletal muscle occur? When it receives signals from (1)
- A. motor neurons that initiates a cascade of actions in muscle fiber.
 - B. sensory neurons that initiates cascade of actions in the muscle fiber.
 - C. interneuron that initiates cascade of actions in the muscle fiber.
 - D. sensory and interneuron that initiates cascade of actions in the muscle fiber.
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12. Which one of the following statements is correct about flat bone? It is part of the (1)
- A. vertebrae that support and maintain the body shape.
 - B. cranium and protect the brain from damage.
 - C. femur and provides structural support to the body.
 - D. mandible that support food chewing in the mouth.
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13. Which of the following is correct about feedback control mechanism during menstrual cycle? Positive feedback mechanism (1)
- A. inhibit the production of progesterone, whereas negative feedback mechanism stimulate the release of progesterone.
 - B. initiated when the volume of gonadotropin reduces, whereas negative feedback triggered with an increased level of gonadotropin.
 - C. acts by reducing LH and FSH to balance other hormones, whereas negative feedback initiates production of more LH and FSH.
 - D. stimulates follicular growth and results in ovulation, whereas negative feedback leads to menstruation.

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- 14.** Which of the following is NOT the consequence of alcoholism in human? It affects the (1)
- A. brain that leads to depression.
 - B. reproductive parts that leads to infertility.
 - C. intestine which leads to colon cancer.
 - D. sensory organs that initiate easy communication.
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- 15.** Carrying capacity is defined as the (1)
- A. place where organisms live and play its role in the environment.
 - B. maximum population that the available resources can sustain.
 - C. growth of population under ideal environmental condition.
 - D. ability of an individuals to cope up a changing environment.
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- 16.** In a given park, there were 237 cheetahs on December 30, 2021. The cheetah population was 309 individuals on December 30, 2024 in the park. What is the average annual growth rate (r) of cheetah population during the given period? (1)
- A. 0.3
 - B. 0.9
 - C. 0.6
 - D. 0.1
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- 17.** What are renewable natural resources? They are resources that (1)
- A. can't be exhausted after repeated utilization.
 - B. are noticeably affected by human consumption.
 - C. have rapid recovery rates after consumption.
 - D. can't naturally replenish in the ecosystem.
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- 18.** Why afforestation is considered as the best soil conservation mechanism? Because it (1)
- A. covers the surface soil from the effect of wind and water.
 - B. facilitates the movement of water during rainy season.
 - C. facilitates the leaching of surface soil minerals.
 - D. reduces humus content of the top soil.
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- 19.** Which one of the following statements is the effect of human activities on the environment? (1)
- A. Releasing water vapor from agricultural and industrial activities
 - B. Releasing CO₂ to atmosphere from coal and natural gas burning
 - C. Minimizing nitrogen contents that plants fix from atmosphere
 - D. Releasing methane gas to the atmosphere from factories
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- 20.** Indigenous knowledge is best defined as a body of knowledge that is built by (1)
- A. searching facts and scientific evidences from various sources.
 - B. group of local people living in a particular area over time.
 - C. developing theories using tested hypothesis from empirical data.
 - D. group of people through standard of experimentation in the field.
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- 21.** What is genetically modified organism means? It is an organism (1)
- A. which maintain species uniformity.
 - B. with high productivity by its nature.
 - C. hybridized using a traditional system of breeding.
 - D. that received hereditary material from other species.

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- 22.** Why waste water from domestic or industrial sources are treated biologically before dumping to the main water body? It is because the process (1)
- A. increases the volume of water and facilitates its evaporation.
 - B. eliminates any microorganisms from the waste water.
 - C. reduces the amount of organic matter of the waste water.
 - D. increases the biochemical oxygen demand of the waste water.
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- 23.** Which of the following is correct about the role of biology in the industry? (1)
- A. Protecting insects from damaging crops
 - B. Producing genetically modified organisms
 - C. Decomposing waste materials during composting
 - D. Producing enzymes through commercial fermentation
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- 24.** Biological warfare is described as the (1)
- A. act of using animals to transport weapons in war.
 - B. applying biological tools to save human life in war.
 - C. act of using biological toxins or infectious agents to kill human.
 - D. production of industrial bio-enzymes for cleaning war equipment.
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- 25.** This item is based on the uses of microorganisms explained below. (1)
- I. Microbial fermentation increases the quality of yogurt and cheese by producing lactic acid.
 - II. Microbes in form of vector inserted into human body to cure defected gene.
 - III. Some bacteria and yeast are able to synthesize insulin at commercial scale.
 - IV. Many bacteria involved in nutrient recycling due to their decomposition role.
- Based on the above list, which of the following order summarize the use of microorganisms in industry, ecosystem, food and medicine, respectively?
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|-------------------|-------------------|
| A. III, IV, II, I | B. I, IV, II, III |
| C. III, IV, I, II | D. I, IV, III, II |
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- 26.** Which one of the following life processes is conducted in living organism to produce larger molecules from smaller ones? (1)
- | | |
|----------------|------------------|
| A. Respiration | B. Decomposition |
| C. Catabolism | D. Anabolism |
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- 27.** Which one of the following statements is correctly describes thylakoid? It is a structure that contains (1)
- A. chlorophyll that serves as a site of light-independent stage of photosynthesis takes place.
 - B. fluid where light-dependent stage of photosynthesis takes place.
 - C. chlorophyll where light-dependent stage of photosynthesis takes place.
 - D. enzyme where light-independent stage of photosynthesis takes place.
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- 28.** Which of the following is correct about chlorophyll a? It is responsible for (1)
- A. absorption of light and giving colors for fruits and seeds.
 - B. collection of light for other photosynthesis pigments.
 - C. indirect transfer of sunlight energy to the photosynthetic pathway.
 - D. absorption and conversion of light energy into chemical energy.

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- 35.** Which one of the following statements is correctly compares the mechanism of water balance during high and low water in human body? The body removes excess water by (1)
- A. producing more urine whereas it conserves water by producing very little urine.
 - B. producing concentrated urine whereas it saves water by reducing evaporation.
 - C. reducing sweating whereas it increases body water by drinking to much water.
 - D. increasing evaporation by sweating whereas it increases body water by producing dilute urine.
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- 36.** It is obvious that anthropogenic and natural factors are the two driving causes of climate change. In what way do these factors aggravate this problem? By releasing greenhouse gases (GHG) during (1)
- A. volcanic eruption in anthropogenic and by flooding in natural factors.
 - B. utilization of biogas in anthropogenic and by wind erosion in natural factors.
 - C. utilization of hydroelectric power in anthropogenic and by earth quick in natural factors.
 - D. burning of fossil fuel in anthropogenic and by volcanic eruption' in natural factors.
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- 37.** How do current climate changes reduce agricultural productivity? By (1)
- A. reducing water availability and increasing temperature
 - B. lowering temperature and annual precipitation of the areas
 - C. reducing harvesting time for agricultural crops
 - D. increasing agricultural crop diversity of an area
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- 38.** Which one of the following statements is INCORRECT about the Kyoto Protocol on climate change? It is an international agreement that (1)
- A. enforces industrialized nations to reduce their GHG emission.
 - B. adopted to eradicate poverty from all UN member states.
 - C. offer market option to countries to meet reduction in GHG emission.
 - D. support low carbon project investment in developing countries.

— END OF EXAMINATION —